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Title

Quantum Computing Notes

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1 An Introduction to Quantum Computing

1.1 Debunking Quantum Computing Myths

- **Myth 1: Quantum computers will solve NP-complete problems efficiently.**
  - *Fact*: False, No efficient classical or quantum algorithm exists for NP-complete problems, and radical theoretical breakthroughs would be required to change this.
- **Myth 2: Quantum computers are too complicated to ever be built.**
  - *Fact*: False, Working Noisy Intermediate-Scale Quantum (NISQ) computers are operational today (e.g., systems built by IBM, Quantinuum, QuEra), including IBM's 1,121-qubit processor.
- **Myth 3: Quantum computing is just another fast hardware architecture (like GPUs or multi-core CPUs).**
  - *Fact*: False, It exploits quantum phenomena with no classical equivalent that cannot be simulated efficiently on classical machines, operating under a fundamentally different model of computation.
- **Myth 4: Quantum computing is hard (and requires knowing quantum physics).**
  - *Fact*: Partially true, but the mathematical foundation (formalized by John von Neumann) relies on complex linear algebra; the main current challenge is building proper high-level programming abstractions.
- **Myth 5: Quantum computers and classical computers solve the same problems.**
  - *Fact*: True and False, in terms of computability (they can simulate each other); however, classical simulation of quantum physics is exponentially slow, whereas quantum algorithms offer exponential speedups for specific tasks.

1.2 Brief History of Quantum Computing

- **1900**: Max Planck studies black-body radiation and discovers that electromagnetic radiation energy is emitted in discrete "quanta".
- **1905**: Albert Einstein explains the photoelectric effect by proposing that light propagates as massless particles called photons.

- **1926**: Erwin Schrödinger formulates the time-dependent ($i\hbar\frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$) and time-independent ($\hat{H}|\Psi\rangle = E|\Psi\rangle$) equations describing the evolution of isolated quantum systems.
- **1927**: Werner Heisenberg introduces the uncertainty principle ($\sigma_x\sigma_p \geq \frac{\hbar}{2}$).
- **1932**: John von Neumann publishes *Mathematical Foundations of Quantum Mechanics*, establishing the theoretical foundation for quantum computing.
- **1963 / 1973**: Yves Lecerf and Charles Bennett introduce reversible Turing machines, proving that isolated quantum systems must undergo reversible computation.
- **1980 / 1982**: Paul Benioff, Yuri Manin, and Richard Feynman propose the first concepts of quantum computers to simulate physical quantum systems.
- **1994**: Peter Shor formulates the quantum factoring algorithm providing an exponential speedup over classical algorithms.
- **1996**: Lov Grover develops the quantum search algorithm providing a quadratic speedup over classical search.

1.3 Current State of the Industry

1.3.1 Sector Investments

- **Public Sector**: Substantial investments in quantum technologies, including the UK (∼£1bn, highlighted in AUKUS), the EU (€1bn Quantum Flagship), the US, Germany/Bavaria (hundreds of millions of euros), and China (field leader in scientific journal publications).
- **Private Sector**: Commercial hardware developers selling systems (IBM, IonQ, IQM, Pasqal, QuEra, Quantinuum), private hardware initiatives (Google, Intel), research teams (Microsoft), cloud integrations (Amazon at Caltech), and financial applications in Quantum Machine Learning and simulation.

1.3.2 Technological Achievements

- **October 2019 (Google)**: Demonstrated quantum supremacy using the 54-qubit *Sycamore* processor.
- **2023 (IBM)**: Made 156-qubit systems publicly available and introduced the 1,121-qubit chip.
- **December 2023 (Harvard/MIT/QuEra)**: Demonstrated error-corrected quantum algorithms on 48 logical qubits created from ∼300 physical qubits

- **April 2024 (Microsoft & Quantinuum)**: Created 4 highly reliable logical qubits from 30 physical qubits, achieving ∼99.8% fidelity and an 800x reduction in error rate.
- **December 2024 (Google)**: Achieved quantum error correction below the surface code threshold on the 105-qubit *Willow* system (executed random circuit sampling benchmark in under 5 minutes vs 10^{25} years required by supercomputers).
- **February 2025 (Microsoft)**: Announced the *Majorana 1* topoconductor chip utilizing Majorana particles for fault-tolerant quantum computing.
- **September 2025 (Harvard)**: Neutral atom progress enabling the creation of 30,000 initialized qubits per second and coherent storage of 3,000 qubits for 2 hours.

2 A review of complex linear algebra

2.1 Why Complex Numbers

The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ represents the most fundamental mathematical object, but it presents immediate limitations: a simple equation such as $x + 21 = 7$ has no solution in $\mathbb{N}$. By allowing negative numbers, the structure is extended to the set of integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, where every equation of the form $x + a = b$ always has the integer solution $x = b - a$ (in our case $x = 7 - 21 = -14$).

Considering an equation like $22x+7 = 0$, one notices that in general there are no integer solutions. It therefore becomes necessary to introduce the set of rational numbers $\mathbb{Q} = \{\dots, -\frac{2}{3}, -2, \dots, -\frac{1}{3}, -\frac{1}{2}, -1, 0, 1, \frac{1}{2}, \frac{1}{3}, \dots, 2, \frac{2}{3}, \dots\}$, in which the general equation $ax+b = 0$ (with $a \neq 0$) admits the rational solution $x = -b/a$ (in this specific case, $x = -7/22$).

Moving on to quadratic equations $ax^2 + bx + c = 0$ ($a \neq 0$ with a, b, c integers), the solutions are given by the formula $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$, valid provided that the discriminant $\Delta = b^2 - 4ac$ is non-negative ($\Delta \geq 0$). If Δ is the square of an integer, the solutions are rational; otherwise, irrational values may emerge (such as $\sqrt{2}$). When $\Delta < 0$, however, one obtains the square root of a negative number. For example, $\sqrt{-4} = \sqrt{4}\sqrt{-1} = 2\sqrt{-1} = 2i$, where $i = \sqrt{-1}$ defines the imaginary unit (such that $i^2 = -1$). A complex number is expressed in the form $a+ib$, with a and b real numbers (an example is $3.17 - 5i$).

When faced with higher-degree polynomials, such as $4x^5 + x^4 + 3.44x^2 + x - 7 = 0$, one might ask whether additional number sets are needed. The Fundamental Theorem of Algebra guarantees that complex numbers are sufficient: every equation of the form $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0$ (with $a_n \neq 0$ and all coefficients a_i complex) admits exactly n complex solutions.

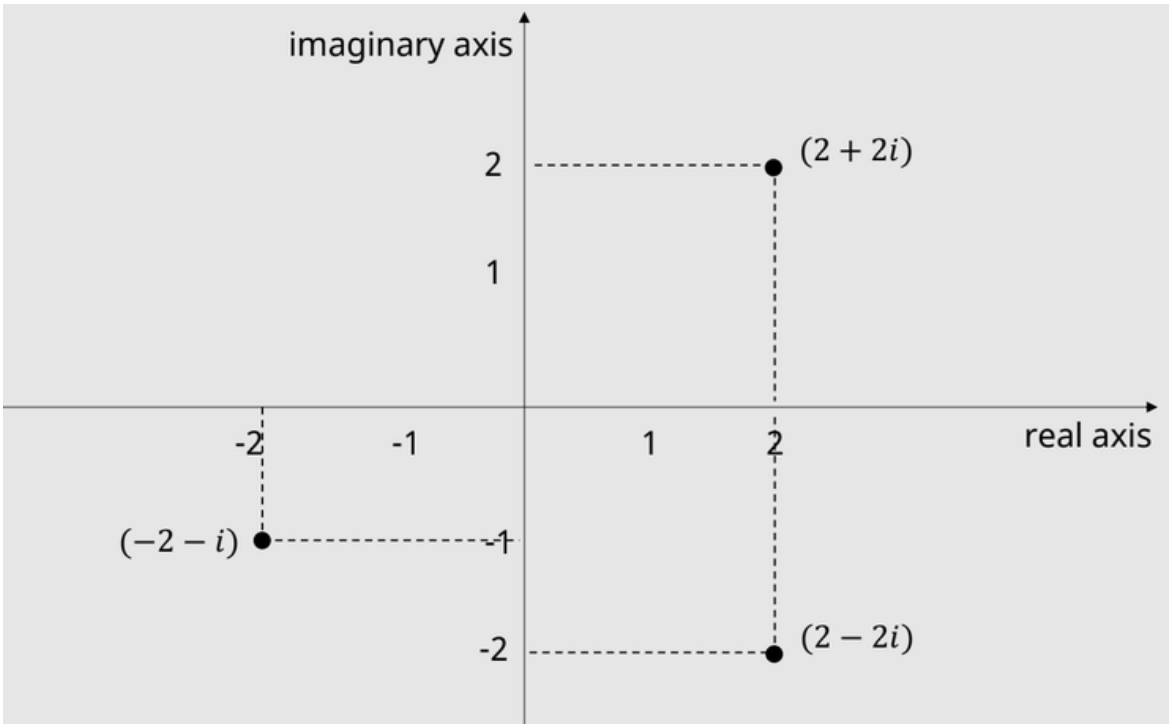
Complex Numbers and Basic Operations

The set of complex numbers is denoted by $\mathbb{C} = \{a+ib \mid a, b \in \mathbb{R}\}$. Given a complex number $z = a + ib$, its real part is $\mathrm{Re}(z) = a$, while its imaginary part is $\mathrm{Im}(z) = b$. Note that the imaginary part of a complex number is a real number!

Complex numbers can be represented on the complex plane (or Argand-Gauss plane), where the horizontal axis measures the real part and the vertical axis measures the imaginary part. Points like $(2 + 2i)$, $(-2 - i)$, and $(2 - 2i)$ are placed precisely as orthogonal coordinate pairs on these axes.

Basic algebraic operations are defined as follows:

- **Addition**: $(a + ib) + (c + id) = (a + c) + i(b + d)$.
- **Multiplication**: $(a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc)$.
- **Conjugate**: The conjugate of $z = a + ib$ is $\overline{z} = a - ib$.



- **Modulus**: The modulus of $z = a + ib$ is the real number $|z| = \sqrt{a^2 + b^2}$, representing the distance of the element from the origin of the Cartesian plane. The relations $|zw| = |z||w|$ and $|z| = \sqrt{z\overline{z}}$ hold.
- **Division**: The division $\frac{z}{w}$ is calculated by multiplying both the numerator and the denominator by the conjugate of the denominator, $\overline{w}$.

2.2 Polar Coordinates and Exponential Form

A generic non-zero complex number $z = a + ib$ with modulus $|z| = \sqrt{a^2 + b^2}$ uniquely determines an angle $\theta \in (-\pi, \pi]$ such that $\cos\theta = \frac{a}{|z|}$ and $\sin\theta = \frac{b}{|z|}$. This allows writing z in polar coordinates as $z = |z|(\cos\theta + i\sin\theta)$. For example, $(2 + 2i) = |2 + 2i|(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) = 2\sqrt{2}(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}})$.

Through the imaginary exponential defined by Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$, one obtains the exponential form $z = |z|e^{i\theta}$. This notation simplifies calculations with powers:

- $z^2 = (|z|e^{i\theta})^2 = |z|^2 e^{i2\theta}$
- $\frac{1}{z} = z^{-1} = (|z|e^{i\theta})^{-1} = |z|^{-1}e^{-i\theta}$
- $\sqrt{z} = (|z|e^{i\theta})^{1/2} = \sqrt{|z|}e^{i\frac{\theta}{2}}$ (showing that the square root always exists in the complex field).

2.3 Vectors and Matrices

A complex vector is an ordered list of complex numbers. Given vectors $v = \begin{pmatrix} 4 \\ i \\ 2+3i \end{pmatrix}$ and $w = \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}$, their sum is performed element by element, yielding $v+w = \begin{pmatrix} 4 \\ 1+i \\ 2+4i \end{pmatrix}$. Scalar multiplication also acts on each element: $v \cdot 2i = \begin{pmatrix} 8i \\ -2 \\ -6+4i \end{pmatrix}$.

The norm of a vector $v = (v_1, v_2, \ldots, v_n)$ expresses its size or amplitude and is defined as $\|v\| = \sqrt{|v_1|^2 + |v_2|^2 + \cdots + |v_n|^2} = \sqrt{\sum_{i=1}^{n} |v_i|^2}$[cite: 1]. For example, the norm of $(4, i, 2+3i)$ is $\sqrt{|4|^2 + |i|^2 + |2+3i|^2} = \sqrt{16+1+13} = \sqrt{30}$.

Complex matrices are tables of complex numbers equipped with specific multiplication rules[cite: 1]. A matrix with m rows and n columns has dimension $m \times n$; when $m = n$, the matrix is called square. The 2×2 identity matrix is given by $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Addition between two matrices is allowed only if they have the same dimension and occurs by adding corresponding entries. For example, adding $\begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix}$ and $\begin{bmatrix} 1 & i \\ i & 10 \\ -4 & 2-3i \end{bmatrix}$ yields $\begin{bmatrix} 4 & 2i \\ 1+2i & 10 \\ -8 & 4 \end{bmatrix}$. Matrix addition enjoys the commutative property: $A + B = B + A$.

Row-by-column product between two matrices A (dimension $m \times n$) and B (dimension $p \times q$) requires $n = p$, producing a matrix of dimension $m \times q$. Multiplying a 3×2 matrix by a 2×2 matrix yields a 3×2 matrix; similarly, a 3×2 matrix multiplied by a 2×1 column vector returns a 3×1 vector[cite: 1]. Scalar multiplication simply multiplies every entry of the matrix by that scalar.

In general, matrix multiplication is NOT commutative ($A \cdot B \neq B \cdot A$). Considering matrices $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, the product $A \cdot B$ generates $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, while $B \cdot A$ produces $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$[cite: 1]. This non-commutative property is of fundamental importance in quantum physics and quantum computing.

2.4 Eigenvectors and Eigenvalues

A non-zero vector v is an eigenvector of a matrix A with eigenvalue λ if it satisfies $A \cdot v = \lambda \cdot v$. Equivalently, this corresponds to $(A - \lambda I) \cdot v = 0$, which admits non-zero solutions $v \neq 0$ if and only if the determinant is zero: $\det(A - \lambda I) = 0$.

As an example, the matrix $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ has eigenvalues $+1$ and -1, with associated eigenvectors $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ respectively. Indeed, for eigenvalue $+1$, verifying directly yields $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i^2 \\ i \end{bmatrix} = \begin{bmatrix} 1 \\ i \end{bmatrix}$.

In general, both eigenvectors v and eigenvalues λ can belong to the complex field, and it is possible for an eigenvalue to be associated with multiple eigenvectors (which form a linear subspace). In quantum mechanics, physical measurements are modeled by specific linear operators (represented by matrices) that always have real eigenvalues.

3 Qubit

3.1 Probabilistic Algorithms

Most standard computer algorithms are **deterministic**: for a given input, they will always produce the exact same output (assuming the program does not crash or hang). Note that multiple distinct input values (e.g., a, b, c) can map to the same output value, which is standard behavior (such as in a simple calculator program).

In contrast, **stochastic (probabilistic) algorithms** exhibit non-deterministic behavior. For a given input i, the program can produce different outputs with specific probabilities:

$$\text{input}_i \longrightarrow \begin{cases} \text{output}_{1,i} & \text{with probability } p_{1,i} \\ \text{output}_{2,i} & \text{with probability } p_{2,i} \\ \text{output}_{3,i} & \text{with probability } p_{3,i} \\ \dots & \\ \text{output}_{N,i} & \text{with probability } p_{N,i} \end{cases}$$

We assume that the number of possible outputs for any input is finite (N), although the set of potential inputs does not need to be finite. For any input i, the probabilities of all possible outputs must sum to 1:

$$\sum_{j=1}^{N} p_{j,i} = 1$$

This condition ensures probabilistic termination, meaning the program will output something with probability 1.

3.2 Classical vs. Quantum Randomness

In classical (non-quantum) computing, probabilistic behavior is introduced artificially using **Pseudo-Random Number Generators (PRNGs)**. PRNGs produce deterministic sequences of numbers designed such that, given one element, predicting the next is computationally difficult. However, because PRNGs rely on finite internal states, their generated sequences are inherently finite and eventually repeat, losing true randomness.

Quantum algorithms, on the other hand, are fundamentally probabilistic by the laws of physics. Quantum systems do not rely on PRNGs; their randomness is intrinsic to the nature of quantum measurements.

3.3 The Quantum Bit (Qubit)

A classical bit is binary and can take a value of either 0 or 1. The quantum equivalent basis states are represented by two-dimensional column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The fundamental feature of a qubit is that its state can exist in a **superposition** of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Sum of the Resulting Vectors

The sum of two vectors is performed by adding the corresponding elements in the same position (row by row):

$$\begin{pmatrix} \alpha \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha + 0 \\ 0 + \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ are complex numbers known as **probability amplitudes**. They satisfy the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

An important example of a qubit state in equal superposition is:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

3.4 Qubit Measurement and Global Phase

The "true" quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ cannot be observed directly. In principle, it is impossible to extract the exact values of the complex amplitudes α and β through a single measurement.

When a qubit is measured in the computational basis, the superposition collapses probabilistically:

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{Measurement}} \begin{cases} |0\rangle & \text{with probability } |\alpha|^2 \\ |1\rangle & \text{with probability } |\beta|^2 \end{cases}$$

Immediately after measurement, the qubit state collapses completely into either $|0\rangle$ or $|1\rangle$.

3.5 Unobservability of Global Phase

For any complex number z with unit modulus ($|z| = 1$), and any complex number α, we have $|z\alpha| = |\alpha|$. Consequently, two qubit states that differ only by a overall complex phase factor z:

$$z\alpha|0\rangle + z\beta|1\rangle \quad \text{and} \quad \alpha|0\rangle + \beta|1\rangle$$

are physically identical and **cannot be distinguished** by any quantum measurement, as the output probabilities remain $|\alpha|^2$ and $|\beta|^2$.

3.6 Qubit Operations

All quantum operations (excluding measurement) are linear transformations. For a general transformation T:

$$T(\alpha|0\rangle + \beta|1\rangle) = \alpha T(|0\rangle) + \beta T(|1\rangle)$$

Every linear transformation on a two-dimensional complex vector space can be represented by a 2×2 matrix.

3.7 Identity and Pauli Matrices

The "do-nothing" transformation is represented by the 2×2 identity matrix:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I|0\rangle = |0\rangle, \quad I|1\rangle = |1\rangle$$

The quantum analogue of the classical NOT gate is the Pauli-X matrix (σ_x):

$$\text{NOT} = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Its action on basis vectors flips the state:

$$\sigma_x|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle, \quad \sigma_x|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

Notice that $\sigma_x^2 = I$.

The other two Pauli matrices widely used in quantum physics and quantum computation are σ_y and σ_z:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Their actions on the computational basis vectors are:

$$\sigma_y|0\rangle = i|1\rangle, \qquad \sigma_y|1\rangle = -i|0\rangle$$
$$\sigma_z|0\rangle = |0\rangle, \qquad \sigma_z|1\rangle = -|1\rangle$$

All Pauli matrices satisfy $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$.

3.8 The Hadamard Transform

The Hadamard gate H is one of the most critical transformations in quantum computing:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Applying H to the basis state $|0\rangle$ creates an equal superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Similarly, applying H to $|1\rangle$ gives:

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

If we apply a measurement to the state $H|0\rangle$, we obtain $|0\rangle$ with probability $1/2$ and $|1\rangle$ with probability $1/2$. Thus, the Hadamard transform applied to a classical basis state acts as a fundamental, truly random bit generator.

3.9 Quantum Evolution and Unitary Matrices

Not every matrix represents a valid quantum transformation. Allowed physical transformations must preserve the norm of the input vector.

Given $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, its norm is $\||\psi\rangle\| = \sqrt{|\alpha|^2 + |\beta|^2} = 1$. Since the total probability of all measurement outcomes must equal 1, quantum transformations must maintain this normalization.

Matrices that preserve vector norms are called **unitary matrices**.

Definition 3.1. A matrix U is unitary if and only if $\|Uv\| = \|v\|$ for any state vector v (or equivalently, $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose of U).

Unitary transformations preserve overall probabilities. It can be easily verified that the Identity matrix, all Pauli matrices $(\sigma_x, \sigma_y, \sigma_z)$, and the Hadamard matrix H are all unitary.

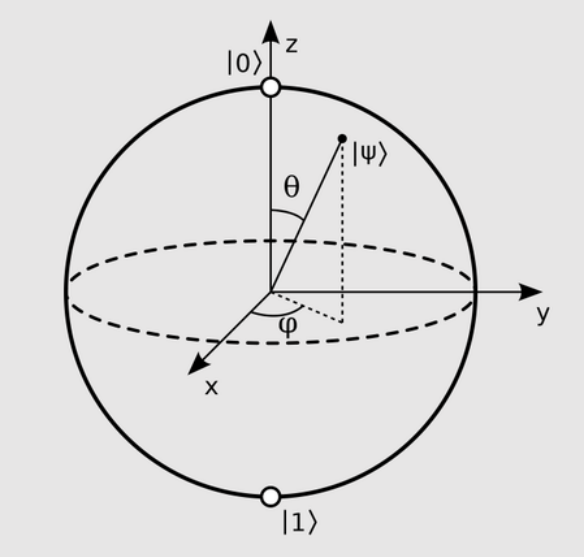
3.10 Picturing Qubits: The Bloch Sphere

Any normalized single-qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ (with $|\alpha|^2 + |\beta|^2 = 1$) can be parameterised using polar coordinates.

Expressing α and β in polar form, the state can be rewritten as:

$$|\psi\rangle = e^{i\gamma} \left(\cos\frac{\theta}{2}|0\rangle + e^{i\varphi} \sin\frac{\theta}{2}|1\rangle \right)$$

where γ, θ, φ are real parameters.



Since the global phase factor $e^{i\gamma}$ has modulus $|e^{i\gamma}| = 1$, it produces no observable physical effects and can be ignored. This simplifies the canonical representation of a single qubit to:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$

where $0 \leq \theta \leq \pi$ and $0 \leq \varphi < 2\pi$.

Geometrically, this parameterization maps state vectors onto the surface of a three-dimensional unit sphere known as the **Bloch Sphere**:

- The North Pole ($\theta = 0$) corresponds to state $|0\rangle$.
- The South Pole ($\theta = \pi$) corresponds to state $|1\rangle$.
- The angle θ represents the polar angle along the z-axis, while φ represents the azimuthal angle in the x-y plane.

Important Note: The Bloch sphere provides an intuitive visual model exclusively for single-qubit states. There is no simple equivalent representation for systems composed of two or more qubits due to quantum entanglement and the exponential growth of the underlying vector space.

4 Multiple Qubit

4.1 Basis Vectors and Classical Bit Systems

Any vector in a vector space can be decomposed as a linear combination of basis vectors. For example:

$$\begin{pmatrix} 3 \\ -2 \\ 1+i \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + (-2) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (1+i) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

This property holds for complex vector spaces of arbitrary dimensions (including infinite dimensions with appropriate conditions).

In general, an n-dimensional vector can be expressed as:

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \alpha_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \cdots + \alpha_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \sum_{i=1}^{n} \alpha_i e_i$$

where e_i is the basis vector with 1 in the i-th entry and 0 elsewhere. These diagonal basis vectors typically correspond to classical states.

4.2 Representation of Multiple Classical Bits

To represent n classical bits, one lists them in a vector or array of n elements. Formally, given the set $\mathcal{B} = \{0, 1\}$, the state space is defined via the Cartesian product $\mathcal{B}^n = \mathcal{B} \times \cdots \times \mathcal{B}$. A classical state is an n-tuple $s = (b_1, b_2, \ldots, b_n) \in \mathcal{B}^n$.

Key characteristics of classical bit representations:

- The length of the vector grows linearly with n.
- Vector components are mutually independent (changing one component does not affect the others).

4.3 Multiple Qubits and Tensor Products

To describe n quantum bits (qubits), the tensor product ($\otimes$) is required.

Single-qubit basis states are defined as $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. A general single-qubit state is a superposition $\alpha_0|0\rangle + \alpha_1|1\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix}$ satisfying $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

4.4 Two-Qubit States

The tensor product of two single-qubit states is computed as:

$$\begin{pmatrix}\alpha_0\\ \alpha_1\end{pmatrix} \otimes \begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix} = \begin{pmatrix}\alpha_0\beta_0\\ \alpha_0\beta_1\\ \alpha_1\beta_0\\ \alpha_1\beta_1\end{pmatrix} = \alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle$$

where $|00\rangle = (1,0,0,0)^T$, $|01\rangle = (0,1,0,0)^T$, $|10\rangle = (0,0,1,0)^T$, and $|11\rangle = (0,0,0,1)^T$.

Why $|00\rangle = (1,0,0,0)^T$

The state $|00\rangle$ is a shorthand notation for the tensor product $|0\rangle \otimes |0\rangle$.

1. Tensor Product (Kronecker Product):

Given the single-qubit state $|0\rangle = \begin{pmatrix}1\\ 0\end{pmatrix}$, we apply the tensor product:

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix}1\\ 0\end{pmatrix} \otimes \begin{pmatrix}1\\ 0\end{pmatrix} = \begin{pmatrix}1 \cdot \begin{pmatrix}1\\ 0\end{pmatrix}\\ 0 \cdot \begin{pmatrix}1\\ 0\end{pmatrix}\end{pmatrix} = \begin{pmatrix}1\cdot 1\\ 1\cdot 0\\ 0\cdot 1\\ 0\cdot 0\end{pmatrix} = \begin{pmatrix}1\\ 0\\ 0\\ 0\end{pmatrix}$$

Written horizontally using the transpose operator T, this becomes $(1,0,0,0)^T$.

2. Binary Ordering:

The four computational basis states correspond to the 2-bit binary representation of decimal integers 0 through 3:

- $|00\rangle \to$ binary $00_2 = 0 \to$ position 1: $(1,0,0,0)^T$
- $|01\rangle \to$ binary $01_2 = 1 \to$ position 2: $(0,1,0,0)^T$
- $|10\rangle \to$ binary $10_2 = 2 \to$ position 3: $(0,0,1,0)^T$
- $|11\rangle \to$ binary $11_2 = 3 \to$ position 4: $(0,0,0,1)^T$

Properties and examples of tensor products for qubits:

- The tensor product extends naturally to any number n of qubits.
- The dimension of the resulting state vector grows exponentially (2^n) with the number of qubits.

- Tensor products yield valid quantum states satisfying normalization: $\sum_{i,j\in\mathcal{B}} |\alpha_i\beta_j|^2 = 1$.
- Example 1: $|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$.
- Example 2: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

The state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ represents an equal superposition over all 2-bit classical states. Although it appears complex, it can be factored into two independent single-qubit states.

4.5 Quantum Entanglement

According to quantum mechanics, any normalized linear combination of the form:

$$\alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle \quad \text{with} \quad \sum_{i=0}^{3} |\alpha_i|^2 = 1$$

is a valid state for a 2-qubit system.

Explanation of the Normalization Condition for a Two-Qubit State

A general state of a two-qubit system is written as a superposition:

$$|\psi\rangle = \alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle$$

The condition $\sum_{i=0}^{3} |\alpha_i|^2 = 1$ is fundamental to quantum mechanics for the following reasons:

1. Superposition Principle:
The four basis states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ form an orthonormal basis for the 4-dimensional complex Hilbert space $\mathbb{C}^4$. Any valid two-qubit state vector $|\psi\rangle$ is a linear combination of these basis vectors with complex coefficients $\alpha_i \in \mathbb{C}$.

2. Born's Rule and Probability Interpretation:
The complex numbers α_i are called *probability amplitudes*. According to Born's Rule, when a measurement is made in the computational basis, the probability $P(i)$ of obtaining outcome i is given by the squared absolute value of its amplitude:

$$P(\text{outcome } i) = |\alpha_i|^2$$

3. Total Probability Conservation:
Since the four basis states represent all mutually exclusive and exhaustive measurement outcomes, the total probability of all possible outcomes combined must equal 1 (100% probability):

$$\sum_{i=0}^{3} P(i) = \sum_{i=0}^{3} |\alpha_i|^2 = |\alpha_0|^2 + |\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 = 1$$

4. Unit Norm Vector in Hilbert Space:
Because the basis vectors are orthonormal ($\langle i|j\rangle = \delta_{ij}$), the inner product of $|\psi\rangle$ with itself yields:

$$\langle\psi|\psi\rangle = \sum_{i=0}^{3} |\alpha_i|^2 = 1$$

This ensures that physical state vectors in quantum mechanics always have a unit norm.

Consider the state $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$:

that is: $|\ \psi\rangle = 0|00\rangle + \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle + 0|11\rangle$

- It cannot be decomposed into two independent local qubit states; it describes the system as an indivisible whole (an *entangled state*).

- It forms an EPR pair (Einstein, Podolsky, Rosen, 1935).
- Measurement of either qubit collapses the system into state $|01\rangle$ with probability $1/2$ or state $|10\rangle$ with probability $1/2$.
- Measuring one qubit immediately reveals the state of the other qubit (which is not the case for separable states like $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$).

Example of a Separable (Divisible) Two-Qubit State

Consider the two-qubit state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

1. Representation in the General Basis:
Written in terms of the general basis $|\psi\rangle = \alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle$:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + 0|01\rangle + \frac{1}{\sqrt{2}}|10\rangle + 0|11\rangle$$

The probability amplitudes are $\alpha_0 = \frac{1}{\sqrt{2}}$, $\alpha_1 = 0$, $\alpha_2 = \frac{1}{\sqrt{2}}$, and $\alpha_3 = 0$.

2. Factorization (Decomposition):
Unlike an entangled state, this state can be factored into a tensor product of two independent single-qubit states $|\psi_A\rangle$ and $|\psi_B\rangle$:

$$|\psi\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle = |+\rangle \otimes |0\rangle$$

- **Qubit A:** $|\psi_A\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$ (equal superposition)
- **Qubit B:** $|\psi_B\rangle = |0\rangle$ (definite state)

Verification via tensor product expansion:

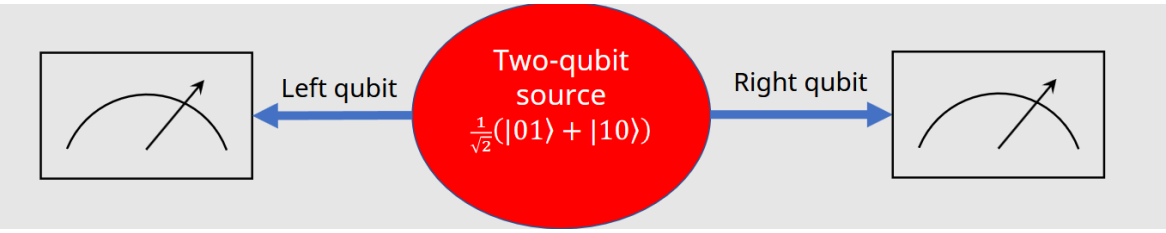
$$\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle) + \frac{1}{\sqrt{2}}(|1\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

3. Key Properties:

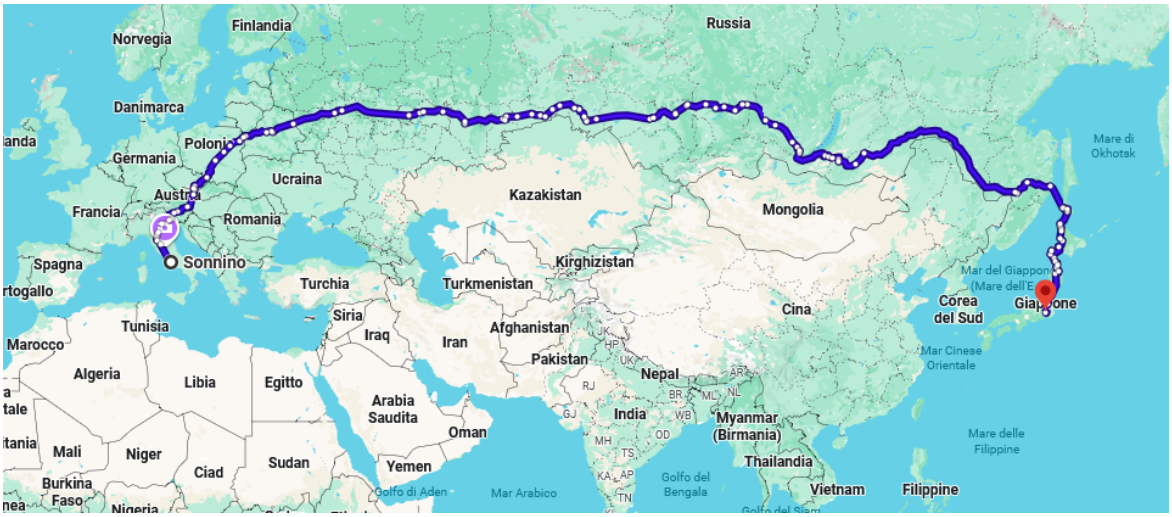
- **Independence:** Each qubit possesses a well-defined individual quantum state.
- **Measurement:** Measuring the first qubit yields $|0\rangle$ with probability 1/2 or $|1\rangle$ with probability 1/2. Measuring the second qubit always yields $|0\rangle$ with 100% certainty, regardless of the first qubit's measurement outcome.
- **No Entanglement:** The measurement of one qubit does not alter or instantaneously determine the state of the other.

4.6 The EPR Paradox and Non-locality

If two entangled qubits fly apart, they maintain their anti-correlation regardless of distance:



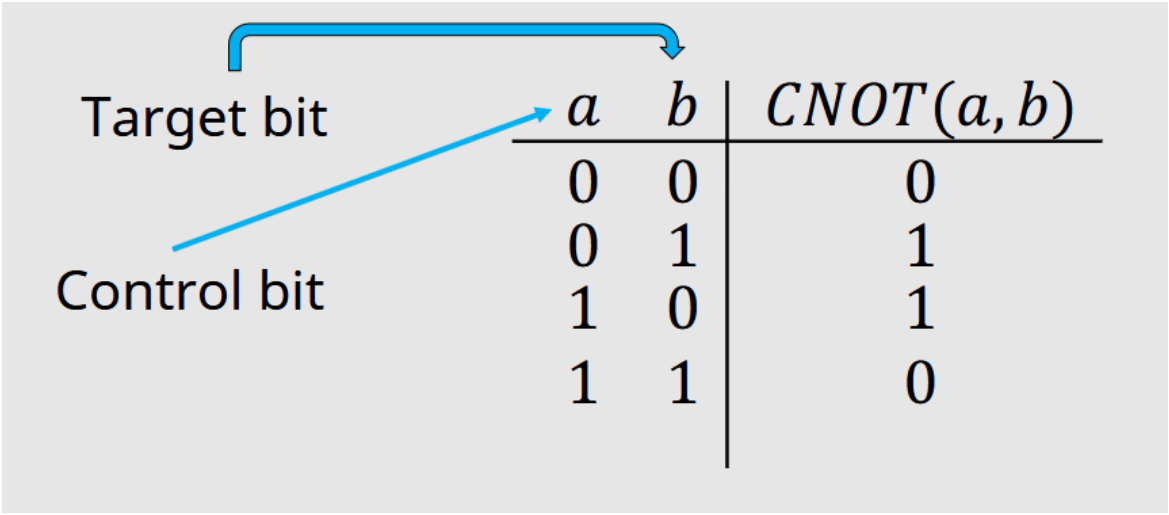
- Einstein referred to this non-local behavior as "spooky action at a distance".
- Entanglement has been experimentally validated over distances exceeding 1,000 km.
- The 2022 Nobel Prize in Physics was awarded to Alain Aspect, John Clauser, and Anton Zeilinger for experiments verifying quantum entanglement.



4.7 The Controlled-NOT (CNOT) Gate

The Controlled-NOT (CNOT) gate operates on two qubits: a control qubit a and a target qubit b.

- Classical functionality: `if a then b := NOT(b) else skip`.



a	b	$CNOT(a,b)$
0	0	0
0	1	1
1	0	1
1	1	0

- Classical truth table: $(0,0) \to (0,0)$, $(0,1) \to (0,1)$, $(1,0) \to (1,1)$, $(1,1) \to (1,0)$.
- Quantum operation: $CNOT(|a,b\rangle) = |a, a \oplus b\rangle$, where $\oplus$ denotes XOR.
- Action on basis states:

$$CNOT(|00\rangle) = |00\rangle, \qquad CNOT(|01\rangle) = |01\rangle,$$
$$CNOT(|10\rangle) = |11\rangle, \qquad CNOT(|11\rangle) = |10\rangle.$$

- 4×4 Matrix representation:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

- Example matrix-vector calculation for $|10\rangle$:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

4.8 Linearity of CNOT and EPR Pair Generation

Since CNOT is a linear operator:

- $CNOT\left(\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$.
- $CNOT\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle\right) = CNOT\left(\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)\right) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ (creates an EPR pair!).

4.9 Quantum Circuit Notation and Multi-Qubit Circuits

Key quantum gates used in circuit diagrams:

- **NOT gate (X)**: Maps $|a\rangle \to |\neg a\rangle$.
- **Hadamard gate (H)**: Maps $|a\rangle \to \frac{1}{\sqrt{2}}\big(|0\rangle + (-1)^a|1\rangle\big)$.
- **CNOT gate**: Maps $|a, b\rangle \to |a, a \oplus b\rangle$.
- **Measurement gate**: Measures $\alpha_0|0\rangle + \alpha_1|1\rangle$, yielding $|0\rangle$ with probability $|\alpha_0|^2$ or $|1\rangle$ with probability $|\alpha_1|^2$.

4.10 Circuits for Entangled States

1. **Creating an EPR Pair**: Starting with $|0\rangle$ (control) and $|1\rangle$ (target), apply an H gate to the control qubit yielding $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$, followed by a CNOT gate to produce the EPR state $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$.
2. **Creating the GHZ State**: The Greenberger–Horne–Zeilinger (GHZ) state is a 3-qubit entangled state. Starting with input $|000\rangle$, apply an H gate to qubit 1, a CNOT gate between qubit 1 (control) and qubit 2 (target), and a CNOT gate between qubit 2 (control) and qubit 3 (target). The resulting state is:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

4.11 Tensor Product of Unitary Gates

For two 2×2 matrices A and B, their tensor product is a 4×4 matrix given by the Kronecker product:

$$A \otimes B = \begin{pmatrix} a_{1,1}B & a_{1,2}B \\ a_{2,1}B & a_{2,2}B \end{pmatrix}$$

Applying $H \otimes H$ to $|00\rangle$ produces the superposition state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

Given $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, the explicit matrix product $H \otimes H$ is calculated as:

$$H \otimes H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1H & 1H \\ 1H & -1H \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

5 Schrödinger's Cat and Macroscopic Entanglement

Quantum mechanical superpositions are routinely observed at the microscopic level (photons, electrons, atoms). However, applying quantum mechanics directly to macroscopic systems leads to conceptual paradoxes.



5.1 The Thought Experiment

Consider a closed box containing:

- A radioactive source whose decay is governed by quantum mechanics.
- A detection device (e.g., a Geiger counter).
- A mechanism connected to a flask of poison and a hammer.
- A cat.

If a radioactive decay occurs, the Geiger counter triggers the hammer, breaking the flask and killing the cat. If no decay occurs, the cat remains alive.

5.2 Quantum Circuit Description

We can model this system using quantum circuits and entangled states:

1. The radioactive source emits a particle initialized in state $|0\rangle$.
2. Applying a Hadamard gate (H) puts the atom into an equal superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

   where $|0\rangle$ denotes "no decay" and $|1\rangle$ denotes "decay".

3. The cat's health state is modeled as a two-level system with basis states $|\text{alive}\rangle = |0\rangle$ and $|\text{dead}\rangle = |1\rangle$.

4. The interaction between the decay and the cat acts as a CNOT (Controlled-NOT) gate, entangling the atom with the cat:

$$|\Psi_{\text{system}}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |\text{alive}\rangle + |1\rangle \otimes |\text{dead}\rangle\right) \equiv \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Upon opening the box (performing a measurement):

- Outcome $|0\rangle \otimes |\text{alive}\rangle$ occurs with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$.
- Outcome $|1\rangle \otimes |\text{dead}\rangle$ occurs with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$.

Macroscopic Superposition and Decoherence

Fuzzy Point in Slides: The slide presents the cat as a simple qubit state $\frac{1}{\sqrt{2}}(|0\otimes\text{alive}\rangle+|1\otimes\text{dead}\rangle)$ without explaining why we do not observe macroscopic superposition in daily life.

Detailed Explanation:

- **Decoherence:** A macroscopic body like a cat consists of $\sim 10^{25}$ particles continuously interacting with thermal photons, air molecules, and environment degrees of freedom ($|E\rangle$).
- The environment acts as an unceasing measurement device, entangling the cat with its surroundings:

$$\frac{1}{\sqrt{2}}|\text{alive}\rangle|E_{\text{alive}}\rangle + \frac{1}{\sqrt{2}}|\text{dead}\rangle|E_{\text{dead}}\rangle$$

Since $\langle E_{\text{alive}}|E_{\text{dead}}\rangle \approx 0$ almost instantaneously (on time scales of 10^{-20} seconds), the quantum interference terms vanish, turning the quantum superposition into a statistical mixture (classical probability).

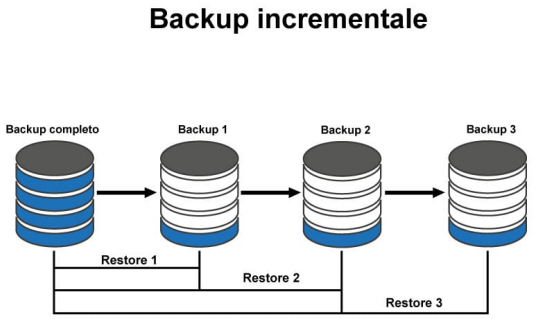
6 The No-Cloning Theorem

Discovered independently by Wootters & Zurek (1982) and Dieks (1982), the No-Cloning Theorem states that **it is impossible to construct a physical quantum operation that creates a duplicate of an arbitrary unknown quantum state**.

6.1 Classical vs. Quantum Copying

In classical computing, a copy operation CP takes a source variable x_i and a blank destination variable y, producing two independent copies:

$$\exists y\ \forall x_i \quad CP(x_i, y) = (x_i, x_i)$$



In quantum mechanics, a copying operator CP must act unitarily on the state space:

$$\forall |x\rangle \quad CP(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |x\rangle$$

where $|y\rangle$ is a fixed target/blank state.

6.2 Proof by Contradiction

Assume such a unitary operator CP exists.

1. Let CP correctly clone two arbitrary quantum states $|x\rangle$ and $|a\rangle$:

$$CP(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |x\rangle$$

$$CP(|a\rangle \otimes |y\rangle) = |a\rangle \otimes |a\rangle$$

2. Consider a linear combination (superposition) state $|x\rangle + |a\rangle$. By the defining property of a cloner, CP must copy this state:

$$CP((|x\rangle + |a\rangle) \otimes |y\rangle) = (|x\rangle + |a\rangle) \otimes (|x\rangle + |a\rangle)$$

3. Expanding the right-hand side via tensor product distributivity:

$$(|x\rangle + |a\rangle) \otimes (|x\rangle + |a\rangle) = (|x\rangle \otimes |x\rangle) + (|x\rangle \otimes |a\rangle) + (|a\rangle \otimes |x\rangle) + (|a\rangle \otimes |a\rangle)$$

Explanation & Proof

The equality

$$(|x\rangle + |a\rangle) \otimes (|x\rangle + |a\rangle) = (|x\rangle \otimes |x\rangle) + (|x\rangle \otimes |a\rangle) + (|a\rangle \otimes |x\rangle) + (|a\rangle \otimes |a\rangle)$$

holds because the **tensor product** ($\otimes$) is a **bilinear operation** over vector spaces (such as Hilbert spaces in quantum mechanics).
Bilinearity guarantees that the tensor product distributes over vector addition from both the left and the right.

Step-by-Step Expansion

1. Right Distributivity
Distribute the right-hand factor $(|x\rangle + |a\rangle)$ over the sum in the left factor:

$$\boxed{(|x\rangle + |a\rangle) \otimes (|x\rangle + |a\rangle) = \Big[(|x\rangle + |a\rangle) \otimes |x\rangle\Big] + \Big[(|x\rangle + |a\rangle) \otimes |a\rangle\Big]}$$

2. Left Distributivity
Apply left distributivity to expand each term individually:

$$\boxed{(|x\rangle + |a\rangle) \otimes |x\rangle = (|x\rangle \otimes |x\rangle) + (|a\rangle \otimes |x\rangle)}$$

$$\boxed{(|x\rangle + |a\rangle) \otimes |a\rangle = (|x\rangle \otimes |a\rangle) + (|a\rangle \otimes |a\rangle)}$$

3. Combining the Terms
Substituting these expansions back into the original expression yields:

$$\boxed{(|x\rangle + |a\rangle) \otimes (|x\rangle + |a\rangle) = (|x\rangle \otimes |x\rangle) + (|x\rangle \otimes |a\rangle) + (|a\rangle \otimes |x\rangle) + (|a\rangle \otimes |a\rangle)}$$

Quantum Mechanics Context

Unlike standard numerical multiplication, the tensor product is **non-commutative**:

$$|x\rangle \otimes |a\rangle \neq |a\rangle \otimes |x\rangle \quad \text{for } |x\rangle \neq |a\rangle$$

The state $|x\rangle \otimes |a\rangle$ indicates that the first subsystem is in state $|x\rangle$ and the second in $|a\rangle$, whereas $|a\rangle \otimes |x\rangle$ represents the reverse. Consequently, the middle terms cannot be combined into $2(|x\rangle \otimes |a\rangle)$ and remain distinct elements in the tensor product space $\mathcal{H}_1 \otimes \mathcal{H}_2$.

4. On the other hand, quantum transformations are **unitary**, and therefore **linear**. Applying CP linearly to $(|x\rangle + |a\rangle) \otimes |y\rangle = |x\rangle \otimes |y\rangle + |a\rangle \otimes |y\rangle$:

$$CP(|x\rangle \otimes |y\rangle + |a\rangle \otimes |y\rangle) = CP(|x\rangle \otimes |y\rangle) + CP(|a\rangle \otimes |y\rangle) = (|x\rangle \otimes |x\rangle) + (|a\rangle \otimes |a\rangle)$$

5. Equating the results from Step 3 and Step 4:

$$(|x\rangle \otimes |x\rangle) + (|a\rangle \otimes |a\rangle) = (|x\rangle \otimes |x\rangle) + (|x\rangle \otimes |a\rangle) + (|a\rangle \otimes |x\rangle) + (|a\rangle \otimes |a\rangle)$$

Subtracting identical terms yields:

$$(|x\rangle \otimes |a\rangle) + (|a\rangle \otimes |x\rangle) = 0$$

This equality is **FALSE** for arbitrary linearly independent non-zero states (e.g., $|x\rangle = |0\rangle$ and $|a\rangle = |1\rangle$ gives $|01\rangle + |10\rangle \neq 0$). Thus, no general quantum cloning operator CP can exist. ■

Alternative Proof via Inner Product Invariance

Fuzzy Point in Slides: The slide proof uses unnormalized superpositions ($|x\rangle + |a\rangle$). Below is the standard inner-product proof showing which states *can* be cloned.

Alternative Rigorous Proof: Let U be a unitary operator that clones two states $|\psi\rangle$ and $|\phi\rangle$:

$$U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle, \quad U(|\phi\rangle \otimes |e\rangle) = |\phi\rangle \otimes |\phi\rangle$$

Taking the inner product of the two transformed states:

$$\Big(\langle\psi| \otimes \langle e|\Big)U^\dagger U\Big(|\phi\rangle \otimes |e\rangle\Big) = \Big(\langle\psi| \otimes \langle\psi|\Big)\Big(|\phi\rangle \otimes |\phi\rangle\Big)$$

Since $U^\dagger U = I$ and $\langle e|e\rangle = 1$:

$$\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2$$

The equation $z = z^2$ for complex numbers $z = \langle\psi|\phi\rangle$ has only two solutions:

- $\langle\psi|\phi\rangle = 0 \implies$ The states are orthogonal.
- $\langle\psi|\phi\rangle = 1 \implies$ The states are identical ($|\psi\rangle = |\phi\rangle$).

Conclusion: A cloning device can only copy states that are mutually orthogonal. It cannot copy an *arbitrary unknown* quantum state.

7 Quantum Teleportation

Quantum teleportation (Bennett et al., 1993) allows Alice to transmit an unknown single-qubit state $|\psi\rangle$ to Bob, situated at a arbitrary distance, using an entangled resource state and 2 bits of classical communication.

7.1 Protocol Setup

- **Input State:** Alice holds an unknown qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$.
- **Shared Resource:** Alice and Bob share an EPR pair (Bell state):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- **Qubit Distribution:**
  - Qubit 1 (q_1): Alice's state to teleport ($|\psi\rangle$).
  - Qubit 2 (q_2): Alice's half of the Bell pair.
  - Qubit 3 (q_3): Bob's half of the Bell pair.

7.2 Step-by-Step Mathematical Derivation

1. Initial Total State

The joint 3-qubit state $|\psi_0\rangle = |q_1\rangle \otimes |q_2\rangle \otimes |q_3\rangle$ is:

$$|\psi_0\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}\left[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)\right]$$

2. Applying CNOT Gate (q_1 as control, q_2 as target)

The CNOT gate acts on the first two qubits (q_1, q_2):

$$\text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|01\rangle = |01\rangle, \quad \text{CNOT}|10\rangle = |11\rangle, \quad \text{CNOT}|11\rangle = |10\rangle$$

Applying this yields $|\psi_1\rangle$:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}\left[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)\right]$$

3. Applying Hadamard Gate (H) on q_1

Recall that $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$|\psi_2\rangle = \frac{1}{2}\left[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)\right]$$

Expanding all terms in the computational basis $|q_1 q_2 q_3\rangle$:

$$|\psi_2\rangle = \frac{1}{2}\left[\alpha(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \beta(|010\rangle + |001\rangle - |110\rangle - |101\rangle)\right]$$

4. Regrouping by Alice's Qubits (q_1, q_2)

Factoring out Alice's 2-qubit basis states $|q_1 q_2\rangle$:

$$|\psi_2\rangle = \frac{1}{2}\left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)\right]$$

5. Measurement and Reconstruction

Alice measures her two qubits (q_1, q_2) in the computational basis, obtaining two classical bits $(b_1, b_2) \in \{0, 1\}^2$. She sends these bits to Bob via a classical channel.

Bob's qubit q_3 collapses into a state depending on (b_1, b_2), which Bob corrects using unitary Pauli operators:

Alice's Result $(b_1 b_2)$	**Bob's Collapsed State** (q_3)	**Bob's Correction Gate**	**Final Recovered State**
00	$\alpha\|0\rangle + \beta\|1\rangle$	Identity (I)	$\alpha\|0\rangle + \beta\|1\rangle = \|\psi\rangle$
01	$\alpha\|1\rangle + \beta\|0\rangle$	Pauli-X (X)	$X(\alpha\|1\rangle + \beta\|0\rangle) = \|\psi\rangle$
10	$\alpha\|0\rangle - \beta\|1\rangle$	Pauli-Z (Z)	$Z(\alpha\|0\rangle - \beta\|1\rangle) = \|\psi\rangle$
11	$\alpha\|1\rangle - \beta\|0\rangle$	Z followed by X (ZX)	$ZX(\alpha\|1\rangle - \beta\|0\rangle) = \|\psi\rangle$

8 Reversible Computation and Deutsch's Algorithm

8.1 Quantum Gates, Unitarity, and Physical Foundations

- **Quantum Gates and Unitarity**: Operations in quantum circuits are represented by complex matrices called **unitary matrices** (U). They satisfy the relation $UU^\dagger = U^\dagger U = I$ and possess the fundamental mathematical property of preserving the length (norm) of state vectors:

$$\|Uv\| = \|v\| \quad \forall v \in \mathcal{H}$$

- **Why Unitarity?**: The physics governing quantum systems is defined by the time-dependent Schrödinger equation:

$$i\hbar \frac{dv(t)}{dt} = Hv(t)$$

  Because the Hamiltonian operator H is self-adjoint ($H = H^\dagger$), the resulting time-evolution operator $U(t_2, t_1) = e^{-\frac{i}{\hbar}H(t_2 - t_1)}$ is strictly unitary. Physically, this guarantees that total probability is conserved over time and never disperses.

- **Reversibility**: Because every quantum gate G is unitary, it always admits a unique inverse gate $G^\dagger$ such that $G^\dagger G = I$, capable of "uncomputing" the operation. As a result, the entire quantum process remains deterministically reversible up until the final measurement.

8.2 Classical vs. Quantum Reversible Computation

Because all quantum gates G are unitary, they are intrinsically reversible:

$$GG^\dagger = G^\dagger G = I$$

This allows any quantum circuit (prior to measurement) to "uncompute" intermediate states.

Classical computation often uses irreversible gates (e.g., AND, OR) that destroy information. However, historical milestones proved that classical computation can be made fully reversible without computational overhead:

- **1963 (Lecerf)**: Formally defined a deterministic, reversible Turing Machine having equivalent computational capability to standard Turing Machines.
- **1973 (Bennett)**: Independently showed that any classical irreversible computation can be embedded into an equivalent reversible computation without losing asymptotic efficiency.
- **2000 (Zuliani)**: Extended reversibility to high-level imperative programming languages incorporating non-determinism and probabilistic semantics ($pGCL$).

8.2.1 The Toffoli Gate (1980)

The Toffoli gate (Controlled-Controlled-NOT) is a 3-bit universal reversible gate defined by:

$$(a, b, c) \mapsto (a', b', c') = (a, b, c \oplus (a \cdot b))$$

Inputs			**Outputs**		
a	b	c	a'	b'	c'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

Emulation of Classical Universality via Toffoli The Toffoli gate transforms input bits (a, b, c) into $(a, b, c \oplus (a \cdot b))$, where $\cdot$ denotes the logical **AND** operation and $\oplus$ denotes the **XOR** operation (addition modulo 2).

- **NAND Emulation**: Set the fixed control input $c = 1$. By leveraging the XOR identity $1 \oplus X = \neg X$ (which acts as a NOT operation):

  $$c' = 1 \oplus (a \cdot b) = \neg(a \cdot b) = \text{NAND}(a, b)$$

  The transformation $(a, b, 1) \mapsto (a, b, \text{NAND}(a, b))$ proves that Toffoli can evaluate any NAND-based classical circuit.

- **FANOUT (Bit Duplication) Emulation**: Set fixed inputs $a = 1$ and $c = 0$ to copy the signal of bit b. Using the identities $1 \cdot b = b$ and $0 \oplus b = b$:

  $$c' = 0 \oplus (1 \cdot b) = 0 \oplus b = b$$

  The transformation $(1, b, 0) \mapsto (1, b, b)$ outputs two identical copies of b on the second and third wires, emulating a deterministic fanout.

8.2.2 High-Level Program Reversibility ($pGCL$)

In $pGCL$, imperative statements S and constructors C are mapped to reversible equivalents (S_r, C_r) and inverses (S_i, C_i):

pGCL Construct S	**Reversible Form** S_r	**Inverse Form** S_i
$v := e$	push v; $v := e$	pop v
skip	skip	skip
$R; S$	$R_r; S_r$	$S_i; R_i$
while c do S od	push b; push F; while c do S_r; push T od	pop b; while b do S_i; pop b od; pop b
$R \triangleleft c \triangleright S$	push b; $(R_r$; push $T) \triangleleft c \triangleright (S_r$; push $F)$	pop b; $(R_i \triangleleft b \triangleright S_i)$; pop b
$R \sqcap S$	push b; $(R_r$; push $T) \sqcap (S_r$; push $F)$	pop b; $(R_i \triangleleft b \triangleright S_i)$; pop b
$R_p \oplus S$	push b; $(R_r$; push $T)_p \oplus (S_r$; push $F)$	pop b; $(R_i \triangleleft b \triangleright S_i)$; pop b

Executing a program, copying its output, and inverting it satisfies:

$$(P_r; P_c; P_i) \equiv (P_c; P)$$

8.3 Quantum Parallelism and the Oracle Model

Given a Boolean function $f : \{0, 1\} \to \{0, 1\}$, its quantum implementation is an oracle U_f:

$$U_f : |x \otimes y\rangle \mapsto |x \otimes (y \oplus f(x))\rangle$$

where $\otimes$ denotes the tensor product and $\oplus$ denotes the **XOR** operation (addition modulo 2).

Evaluating U_f on an equal superposition input state yields:

$$U_f\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\otimes|0\rangle\right)=\frac{1}{\sqrt{2}}\left(|0\otimes f(0)\rangle+|1\otimes f(1)\rangle\right)$$

This evaluates $f(0)$ and $f(1)$ simultaneously in a single application of U_f (**Quantum Parallelism**).

Clarification & Omitted Detail: Step-by-Step Derivation of Quantum Parallelism

To see how this evaluation works in detail, initialize $y=0$ and pass x in equal superposition:

1. **Expand Input State via Tensor Product Distributivity**:

$$\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\otimes|0\rangle=\frac{1}{\sqrt{2}}\left(|0\otimes 0\rangle+|1\otimes 0\rangle\right)$$

2. **Apply U_f by Linearity**:

$$U_f\left(\frac{|0\otimes 0\rangle+|1\otimes 0\rangle}{\sqrt{2}}\right)=\frac{1}{\sqrt{2}}\left(U_f|0\otimes 0\rangle+U_f|1\otimes 0\rangle\right)$$

3. **Apply Oracle Mapping $U_f|x\otimes y\rangle=|x\otimes(y\oplus f(x))\rangle$ with $y=0$**:

$$=\frac{1}{\sqrt{2}}\left(|0\otimes(0\oplus f(0))\rangle+|1\otimes(0\oplus f(1))\rangle\right)$$

4. **Simplify via XOR Identity ($0\oplus X=X$)**:

$$=\frac{1}{\sqrt{2}}\left(|0\otimes f(0)\rangle+|1\otimes f(1)\rangle\right)$$

8.4 Deutsch's Problem and Query Complexity

Deutsch's problem asks whether a given function $f:\{0,1\}\to\{0,1\}$ is **constant** ($f(0)=f(1)$) or **balanced** ($f(0)\neq f(1)$). This is equivalent to computing:

$$f(0)\oplus f(1)$$

- **Classical Complexity**: Requires **2 queries** ($f(0)$ and $f(1)$ must both be evaluated).
- **Quantum Complexity**: Requires **exactly 1 query** using Deutsch's Algorithm.

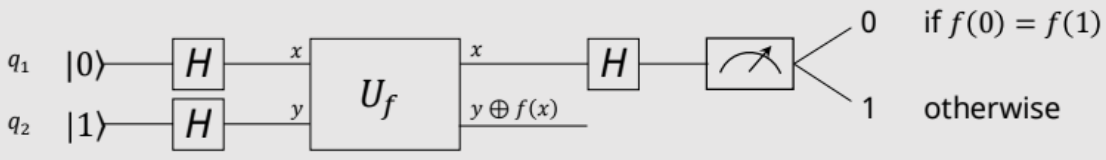
$f(0)$	$f(1)$	$f(0) \oplus f(1)$	**Function Type**
0	0	0	Constant
0	1	1	Balanced
1	0	1	Balanced
1	1	0	Constant

8.5 Deutsch's Algorithm: Circuit and Step-by-Step Derivation

8.5.1 Circuit Diagram and Pseudocode

Using a "programming" notation:

$$q_1, q_2 = |01\rangle;$$
$$q_1, q_2 = H \otimes H(q_1, q_2);$$
$$q_1, q_2 = U_f(q_1, q_2);$$
$$q_1 = H(q_1);$$
$$b = Measure(q_1);$$



Deutsch's Algorithm in 3 Steps

1. **Quantum Parallelism**: The first qubit is initialized in equal superposition:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

   By applying the oracle U_f only once, it computes $f(0)$ and $f(1)$ at the same time in parallel.

2. **Phase Kickback**: The second qubit is initialized in state:

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

The values $f(x)$ are not saved directly in the second qubit; instead, they are kicked back as a phase factor $(-1)^{f(x)}$ in front of the first qubit.

3. **Interference and Measurement**: A final Hadamard gate H is applied to the first qubit before measurement. The superposition branches interfere:
   - If f is **constant**, interference cancels out state $|1\rangle$ and leaves state $|0\rangle$.
   - If f is **balanced**, interference cancels out state $|0\rangle$ and leaves state $|1\rangle$.

Clarification & Omitted Detail: The Main Takeaway

Measuring the first qubit gives a 100% deterministic result:

- Result 0 $\implies$ f is **constant**.
- Result 1 $\implies$ f is **balanced**.

This shows the true advantage of quantum computing: using superposition and interference to compute a global property of a function $(f(0) \oplus f(1))$ in a single oracle query, without needing to know $f(0)$ and $f(1)$ separately.

9 In-Orbit Programmable Quantum Photonic Processors

While ground-based quantum processors have demonstrated computational speedups for specialized tasks, deploying quantum hardware into Low Earth Orbit (LEO) is essential for scalable satellite-based quantum networks. A milestone experiment led by the University of Vienna (Steiner et al., 2026) successfully operated the first reconfigurable, programmable quantum photonic processor aboard a nanosatellite in LEO (altitude $\sim$ 510 km). This mission transitions space quantum technology from passive quantum key distribution (QKD) to active, programmable quantum processing units operating directly in space.

Payload Architecture and Mathematical Model The onboard quantum processor integrates state generation, linear optical routing, and single-photon detection:

- **Single-Photon Source**: A 405 nm pump laser drives a Periodically Poled Potassium Titanyl Phosphate (PPKTP) crystal via Type-II Spontaneous Parametric Down-Conversion (SPDC), generating photon pairs at 810 nm.
- **Universal Photonic Integrated Circuit (uPIC)**: A 6-mode glass chip featuring 15 thermo-optically controlled Mach-Zehnder Interferometers (MZIs).
- **Control and Detection**: Single-Photon Avalanche Diodes (SPADs) paired with an FPGA for real-time coincidence counting.

Mathematically, the 6-mode uPIC implements a linear unitary transformation $U \in SU(6)$ on input optical states:

$$|\psi_{\text{out}}\rangle = U|\psi_{\text{in}}\rangle$$

When two indistinguishable photons enter modes i and j, two-photon interference is governed by the permanent of the submatrix $U_{i,j}$:

$$P(k,l) = |\text{Perm}(U_{i,j\to k,l})|^2$$

Experimental Results and Space Operations The payload underwent an 8-month test campaign in orbit, demonstrating structural resilience and quantum interference:

- **Classical Circuit Fidelity**: Testing 9 unitary configurations with distinguishable photons yielded a mean fidelity of $\bar{F} = 0.888 \pm 0.011$ (reaching $\bar{F} = 0.949 \pm 0.014$ when excluding temperature calibration anomalies).
- **Two-Photon Interference**: Tuning the PPKTP crystal to its degeneracy temperature ($\sim 32.5°$C) allowed the observation of the Hong-Ou-Mandel (HOM) interference dip in LEO, confirming preserved photon indistinguishability.
- **Operational Constraints**: Launch vibrations disabled 3 of the 6 SPAD detectors, restricting tests to functional 3-mode subspaces. Radiation and solar light noise were managed by executing experiments during orbital night.

Clarification & Omitted Detail: Key Findings and Significance

- **Flight Heritage**: Validates that integrated photonic chips and non-linear SPDC sources can survive rocket launch acceleration and LEO space radiation.
- **Onboard Processing**: Enables local quantum pre-processing of space data before classical transmission to ground stations.
- **Orbital Quantum Nodes**: Provides a scalable building block for routing quantum information across future global quantum networks.

10 The Deutsch-Jozsa Algorithm

10.1 The Deutsch-Jozsa Problem

The Deutsch-Jozsa algorithm generalizes Deutsch's algorithm from a single-bit input to an n-bit Boolean function, demonstrating an exponential deterministic query speedup.

Consider an unknown black-box Boolean function (oracle):

$$f : \mathcal{B}^n \to \mathcal{B}, \quad \text{where } \mathcal{B} = \{0,1\}$$

The function maps an n-bit binary string $x = (x_{n-1}, \ldots, x_0) \in \{0,1\}^n$ to a single bit $f(x) \in \{0,1\}$. It is guaranteed to be strictly either:

- **Constant**: $f(x)$ yields the same output value for all inputs ($\forall x \in \mathcal{B}^n, f(x) = 0$ or $\forall x \in \mathcal{B}^n, f(x) = 1$).
- **Balanced**: $f(x)$ yields 0 for exactly half of the possible inputs and 1 for the remaining half:
$$\sum_{x \in \mathcal{B}^n} f(x) = 2^{n-1}$$

Query Complexity Comparison

- **Classical Deterministic Worst-Case**: To determine with 100% certainty whether f is constant or balanced, a classical algorithm must check at most $2^{n-1} + 1$ distinct input values in the worst case.
- **Quantum Complexity**: The Deutsch-Jozsa algorithm determines the nature of f deterministically (100% accuracy) with **exactly 1 evaluation** of U_f.

10.2 The n-Qubit Hadamard Transform

For a single-qubit state $a \in \mathcal{B} = \{0,1\}$, the Hadamard operation can be written compactly as:
$$H|a\rangle = \frac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{a \cdot b} |b\rangle$$

For a two-qubit state $|xy\rangle = |x\rangle \otimes |y\rangle$ with $x, y \in \mathcal{B}$:

$$\begin{aligned}(H \otimes H)|xy\rangle &= \left(\frac{1}{\sqrt{2}} \sum_{a \in \mathcal{B}} (-1)^{x \cdot a} |a\rangle \right) \otimes \left(\frac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{y \cdot b} |b\rangle \right) \\ &= \frac{1}{2} \sum_{a,b \in \mathcal{B}} (-1)^{x \cdot a + y \cdot b} |a\rangle|b\rangle = \frac{1}{2} \sum_{a \in \mathcal{B}^2} (-1)^{xy \cdot a} |a\rangle\end{aligned}$$

By extension, applying $H^{\otimes n}$ to an n-qubit basis state $|x\rangle$ with $x \in \mathcal{B}^n$ yields:
$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle$$

where $x \cdot a$ denotes the bitwise inner product modulo 2:
$$x \cdot a = \bigoplus_{i=0}^{n-1} x_i a_i = \left(\sum_{i=0}^{n-1} x_i a_i \right) \pmod 2$$

10.3 Algorithm Execution and Step-by-Step Derivation

The algorithm employs two quantum registers:

- q_1: An n-qubit input register initialized to $|0^n\rangle$.
- q_2: A 1-qubit target register initialized to $|1\rangle$.

The operational sequence follows the execution steps:

$$\begin{aligned}
&q_1, q_2 = |0^n 1\rangle; \\
&q_1, q_2 = (H^{\otimes n} \otimes H)(q_1, q_2); \\
&q_1, q_2 = U_f(q_1, q_2); \\
&q_1 = H^{\otimes n}(q_1); \\
&b = \text{Measure}(q_1);
\end{aligned}$$

Step 1: System Initialization The initial state of the joint system is:

$$|\psi_0\rangle = |0^n\rangle \otimes |1\rangle = |0^n 1\rangle$$

Step 2: Superposition Generation ($H^{\otimes n} \otimes H$) Applying $H^{\otimes n}$ to q_1 and H to q_2 produces:

$$|\psi_1\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} |x\rangle \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Step 3: Oracle Application (U_f) and Phase Kickback The oracle U_f maps $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$. By phase kickback:

$$U_f \left(|x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^{f(x)} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Evaluating this across the equal superposition yields:

$$|\psi_2\rangle = \left(\sum_{x \in \mathcal{B}^n} \frac{(-1)^{f(x)} |x\rangle}{\sqrt{2^n}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Step 4: Final Hadamard Transformation ($H^{\otimes n}$ on q_1) Applying $H^{\otimes n}$ to q_1 maps each state $|x\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle$:

$$\begin{aligned}
|\psi_3\rangle &= (H^{\otimes n} \otimes I)|\psi_2\rangle \\
&= \sum_{x \in \mathcal{B}^n} \frac{(-1)^{f(x)}}{\sqrt{2^n}} \left(\frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
&= \sum_{a \in \mathcal{B}^n} \left(\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)} \right) |a\rangle \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)
\end{aligned}$$

10.4 Measurement Analysis and Decision Rule

Measuring the n-qubit register q_1 in the computational basis $\mathcal{B}^n$ yields state $|a\rangle$ with amplitude:

$$C_a = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)}$$

To evaluate whether f is constant or balanced, analyze the probability amplitude C_{0^n} for state $|0^n\rangle = |00\ldots0\rangle$. Since $x \cdot 0^n = 0$ for all x:

$$C_{0^n} = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)}$$

1. **Case 1: f is Constant**

   - If $f(x) = 0$ for all x, $C_{0^n} = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (+1) = 1$.
   - If $f(x) = 1$ for all x, $C_{0^n} = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1) = -1$.

   In both cases, $|C_{0^n}|^2 = |\pm 1|^2 = 1$. Constructive interference concentrates the entire probability on $|0^n\rangle$, making the probability of measuring any $a \neq 0^n$ strictly zero.

2. **Case 2: f is Balanced** Since $f(x) = 0$ for 2^{n-1} inputs and $f(x) = 1$ for 2^{n-1} inputs, the terms cancel:

   $$C_{0^n} = \frac{1}{2^n} \left(\sum_{x : f(x)=0} (+1) + \sum_{x : f(x)=1} (-1) \right) = \frac{1}{2^n} (2^{n-1} - 2^{n-1}) = 0$$

   Destructive interference reduces the probability of measuring $|0^n\rangle$ to zero ($|C_{0^n}|^2 = 0$). Measurement will yield some non-zero string $a \neq 0^n$.

Decision Rule

- Measuring $|0^n\rangle \implies f$ is **Constant**.
- Measuring any state $a \neq 0^n \implies f$ is **Balanced**.

Clarification & Omitted Detail: Key Takeaways: Deutsch-Jozsa Algorithm

- **Exponential Speedup**: Reduces worst-case query complexity from $O(2^n)$ classical queries to a single quantum query ($O(1)$).
- **Quantum Interference**: Constructive interference selects $|0^n\rangle$ for constant functions, while destructive interference eliminates $|0^n\rangle$ for balanced functions.

- **Global Feature Extraction**: Demonstrates the quantum capacity to evaluate a global structural property of f without determining individual evaluation points $f(x)$.

11 Grover's Quantum Search Algorithm: Complete Lecture Summary

11.1 The Search Problem Formulation

- **Array Configuration**: Consider an unstructured array A of length $N = 2^n$ containing M solution elements, with $1 \leq M \leq N$.
- **Concrete Example**: An array of $N = 16$ elements (indexed 0 to 15) with values:

$$[78, 655, 9797, 3249, 6, 13, 877, 56, 8789, 10, 999, 1548, 354, 75, 1875, 9]$$

- **Goal**: Find an index $i \in \{0, \ldots, N-1\}$ such that $A[i]$ is a solution element.
- **Index Encoding**: Array indices $i \in \{0, \ldots, N-1\}$ are represented using n-bit binary strings.
- **Boolean Solution Function**: Solutions are encoded via a Boolean function $f : \{0, \ldots, N-1\} \to \{0, 1\}$ defined as:

$$f(i) = \begin{cases} 0 & \text{if } A[i] \text{ is not a solution} \\ 1 & \text{if } A[i] \text{ is a solution} \end{cases}$$

- **Complexity Comparison**:
  - *Classical Random Search*: Requires $O\left(\frac{N}{M}\right)$ array accesses in the worst-case scenario.
  - *Quantum (Grover's Search)*: Returns a correct solution index with high probability using only $O\left(\sqrt{\frac{N}{M}}\right)$ array accesses.
- **Big-O Notation Reminder**: $f(x) = O(g(x))$ as $x \to \infty$ if $|f(x)| \leq K|g(x)|$ for some constant $K > 0$ and sufficiently large x.

11.2 Constructing the Quantum Oracle and Phase Kickback

- **Bit-Oracle Unitary Operator** U_f: Given $f : \{0, \ldots, N-1\} \to \{0, 1\}$, we build a unitary operator U_f acting on an n-qubit register $|x\rangle$ and a single target qubit $|y\rangle$:

$$U_f : |x \otimes y\rangle \mapsto |x \otimes (y \oplus f(x))\rangle$$

- **Core Actions of** U_f:

$$U_f|x \otimes 0\rangle = |x \otimes f(x)\rangle \quad \text{and} \quad U_f|x \otimes 1\rangle = |x \otimes \neg f(x)\rangle$$

- **Phase Kickback Mechanism**: By setting the target qubit to the state $|y\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, U_f evaluates as follows:

$$\begin{aligned} U_f\left(|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) &= |x\rangle \otimes \frac{1}{\sqrt{2}}(|f(x)\rangle - |\neg f(x)\rangle) \\ &= \begin{cases} |x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) & \text{if } f(x) = 0 \\ |x\rangle \otimes \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) & \text{if } f(x) = 1 \end{cases} \\ &= (-1)^{f(x)}|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \end{aligned}$$

- **Phase Oracle** O_f: Because the right-hand side target qubit remains completely unaltered, it can be dropped, leaving the phase oracle operator:

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle$$

- **Oracle Paradigm**: O_f flips the phase (sign of the amplitude) of solution states ($f(x) = 1$) while leaving non-solutions ($f(x) = 0$) unchanged. Grover's algorithm is a primary example of oracle (black-box) quantum algorithms, akin to the Deutsch-Jozsa algorithm.

11.3 Grover's Quantum Circuit and Iteration Operator

- **Circuit Architecture**:

  1. *Initialization*: n work qubits are initialized to $|0^n\rangle$ alongside an oracle workspace qubit initialized to $|1\rangle$.
  2. *Hadamard Transformation*: Apply $H^{\otimes n}$ to all n work qubits to create a uniform superposition:

$$|\psi\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle$$

3. *Grover Iterations*: Apply the Grover iteration operator G repeatedly for $O\left(\sqrt{\frac{N}{M}}\right)$ steps.
4. *Measurement*: Measuring the n-qubit register yields a solution index with probability ≈ 1 and a non-solution index with probability ≈ 0.

- **General n-Qubit Quantum State**: Represented as $\sum_{x\in\{0,1\}^n} \alpha_x|x\rangle$ with normalization condition $\sum_{x\in\{0,1\}^n} |\alpha_x|^2 = 1$.

- **Formal Definition of Grover's Operator G**:

$$G = W \cdot O_f$$

where W is the unitary transformation performing *inversion about the mean*:

$$W|x\rangle = (-\alpha_x + 2\langle\alpha\rangle)|x\rangle = (2\langle\alpha\rangle - \alpha_x)|x\rangle$$

with the mean amplitude $\langle\alpha\rangle$ defined by:

$$\langle\alpha\rangle = \frac{1}{N}\sum_{x\in\{0,1\}^n} \alpha_x$$

- **Step-by-Step Dynamics of Grover's Iteration G**:

1. *Initial Uniform State*: All amplitudes are real and equal to $\frac{1}{\sqrt{N}}$.
2. *Oracle Application (O_f)*: Inverts the sign of solution amplitudes to $-\frac{1}{\sqrt{N}}$, which lowers the overall mean amplitude $\langle\alpha\rangle$.
3. *Inversion About the Mean (W)*: Replaces each amplitude α_x with $2\langle\alpha\rangle - \alpha_x$. For non-solutions, amplitudes drop slightly below $\frac{1}{\sqrt{N}}$. For solutions, the negative amplitude flips over the mean to $2\langle\alpha\rangle - \left(-\frac{1}{\sqrt{N}}\right) = 2\langle\alpha\rangle + \frac{1}{\sqrt{N}}$, drastically amplifying its positive value.

11.4 Geometric Interpretation in Two-Dimensional Subspace

- **Subspace Decomposition**: Let $S \subset \{0, \ldots, N-1\}$ be the set of solution indices ($|S| = M$, e.g., $S = \{2, 12\}$ in the 16-element example) and $\overline{S}$ be the set of non-solution indices. Define two orthonormal basis vectors:
  - Superposition of non-solution states: $|a\rangle = \frac{1}{\sqrt{N-M}}\sum_{x\notin S}|x\rangle$
  - Superposition of solution states: $|b\rangle = \frac{1}{\sqrt{M}}\sum_{x\in S}|x\rangle$

- **Initial Uniform Superposition** $|\psi\rangle$: Expressed in the 2D plane spanned by $\{|a\rangle, |b\rangle\}$:

$$|\psi\rangle = \frac{1}{\sqrt{N}}\sum_{x\in\{0,1\}^n}|x\rangle = \sqrt{\frac{N-M}{N}}|a\rangle + \sqrt{\frac{M}{N}}|b\rangle$$

- O_f **as a Reflection**: The oracle reflects any state vector across the non-solution axis $|a\rangle$:

$$O_f(\alpha|a\rangle + \beta|b\rangle) = \alpha|a\rangle - \beta|b\rangle$$

- W **as a Reflection**: Let $P_\psi = |\psi\rangle\langle\psi|$ be the projector onto $|\psi\rangle$. W can be written as:

$$W = 2P_\psi - I = 2P_\psi - (P_\psi + P_{\psi^\perp}) = P_\psi - P_{\psi^\perp}$$

This proves that W performs a reflection across the initial state vector $|\psi\rangle$.

- G **as a Planar Rotation**: Since the composition of two 2D reflections across lines separated by angle $\theta/2$ is a rotation by angle θ, $G = WO_f$ performs a **pure rotation** in the plane spanned by $\{|a\rangle, |b\rangle\}$.

11.5 Angular Dynamics, Iteration Count, and Complexity Proof

- **Angle Definition**: Let $\theta/2$ be the angle between $|a\rangle$ and $|\psi\rangle$. From $|\psi\rangle = \sqrt{\frac{N-M}{N}}|a\rangle + \sqrt{\frac{M}{N}}|b\rangle$:

$$\cos\frac{\theta}{2} = \sqrt{\frac{N-M}{N}} \quad \text{and} \quad \sin\frac{\theta}{2} = \sqrt{\frac{M}{N}}$$

Thus, $|\psi\rangle = \cos\frac{\theta}{2}|a\rangle + \sin\frac{\theta}{2}|b\rangle$.

- **State Evolution After** k **Iterations**: Each application of $G = WO_f$ rotates the state by θ towards $|b\rangle$:

$$G|\psi\rangle = \cos\frac{3\theta}{2}|a\rangle + \sin\frac{3\theta}{2}|b\rangle$$
$$G^k|\psi\rangle = \cos\frac{(2k+1)\theta}{2}|a\rangle + \sin\frac{(2k+1)\theta}{2}|b\rangle \quad \text{for } k = 0, 1, 2, \ldots$$

- **Optimal Iteration Count** k: To maximize the measurement success probability, we must rotate $|\psi\rangle$ as close as possible to $|b\rangle$. The required number of iterations is:

$$k = \left\lfloor \frac{\arccos\sqrt{\frac{M}{N}}}{\theta} \right\rfloor$$

This places the state within an angle of $\leq \theta/2$ from $|b\rangle$, guaranteeing a solution measurement probability of at least 50%.

- **Mathematical Complexity Proof ($O(\sqrt{N/M})$ Upper Bound)**: Assuming $M \leq \frac{N}{2}$ (which ensures $\theta \leq \frac{\pi}{2}$):

1. Since $\arccos(x) \leq \frac{\pi}{2}$ for $x \geq 0$, we have $\arccos\sqrt{\frac{M}{N}} \leq \frac{\pi}{2}$.

2. Using the inequality $\sin x \leq x$ for $x \geq 0$:

$$\sqrt{\frac{M}{N}} = \sin\frac{\theta}{2} \leq \frac{\theta}{2} \implies \frac{1}{\theta} \leq \frac{1}{2}\sqrt{\frac{N}{M}}$$

3. Combining these bounds yields:

$$k = \left\lceil \frac{\arccos\sqrt{\frac{M}{N}}}{\theta} \right\rceil \leq \left\lceil \frac{\pi}{2\theta} \right\rceil \leq \left\lceil \frac{\pi}{4}\sqrt{\frac{N}{M}} \right\rceil = O\left(\sqrt{\frac{N}{M}}\right)$$

- **Concluding Open Question**: How can we remove the constraint $M \leq \frac{N}{2}$?

References

[1] Leslie Lamport. *LaTeX: a Document Preparation System.* Addison Wesley, Massachusetts, 2 edition, 1994.